

DATA ANALYSIS USING PYTHON

4-Week Practical Learning & Industry Project Submission

Intern Name: Abdul Haq

Program Duration: 4 Weeks (Remote)

Domain: Data Analytics & Business Intelligence

Dataset: Netflix Global Titles Dataset

Tools Used: Python, Pandas, Matplotlib, Seaborn

Submission Date: October 2026

1. Executive Summary & Program Overview

This report presents the complete execution and analytical findings of the **Data Analysis Using Python Internship** conducted by **Auspify Technologies**. The program focuses on bridging academic learning and industry execution through structured project-based training.

The primary objective of this project is to perform end-to-end exploratory data analysis (EDA), data cleaning, content categorisation, trend extraction, and business intelligence reporting on the global **Netflix Dataset** containing **8,790 catalog items**.

2. Dataset Overview & Data Quality Assessment

8,790

TOTAL RECORDS

10

ATTRIBUTES / COLUMNS

0

MISSING CELL VALUES

100%

DATA COMPLETENESS

Attribute Name	Data Type	Sample Entry	Quality & Integrity Status
show_id	String / Identifier	s1, s2, s3 ...	100% Unique, zero duplicates
type	Categorical	Movie, TV Show	Standardized (2 unique types)
title	Text	Dick Johnson Is Dead	Clean text strings
director	Text / List	Kirsten Johnson	Contains 'Not Given' placeholders
country	Categorical / Geographic	United States	287 'Not Given' values isolated
date_added	Date / String	September 25, 2021	Parsed to YYYY-MM-DD datetime
release_year	Integer	2020	Valid numerical range (1925–2021)
rating	Categorical	PG-13, TV-MA	Grouped into standard age tiers

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DATA ANALYTICS TASK EXECUTION • TASKS 1 & 2

DATA CLEANING & DASHBOARD

Task 1 EASY - Netflix Data Cleaning & Preparation

Objective: Prepare, standardize, and structure the raw Netflix dataset for downstream business analytics and statistical visualization.

✓ Execution Workflow & Methodology

1. Imported raw dataset using Python pandas .read_csv() and verified structural integrity.
2. Scanned for missing null values and identified categorical placeholders like 'Not Given'.
3. Standardized categorical text fields (stripped extra whitespaces, formatted text casing).
4. Converted the date_added string values into structured Python Datetime objects.
5. Exported clean data baseline ready for multi-dimensional business analytics.

```
# Python Code Implementation Snippet - Task 1
import pandas as pd # Step 1 & 2: Load Data & Inspect Quality
df = pd.read_csv('Dataset.csv')
df['date_added'] = pd.to_datetime(df['date_added'].str.strip(), format='%B %d, %Y', errors='coerce') # Step 3 & 4: Standardize Placeholders & Text Strings
df['country'] = df['country'].replace('Not Given', 'Unspecified')
df['rating'] = df['rating'].str.strip()
print(f"Cleaned Dataset Shape: {df.shape} | Null Count: {df.isnull().sum().sum()}")
```

Task 2 EASY - Content Type Analysis Dashboard

Objective: Quantify and analyze the distribution breakdown between Movies and TV Shows across the global Netflix platform.

Content Type	Total Count	Catalog Share (%)
Movies	6,126	69.69%
TV Shows	2,664	30.31%
Total Catalog	8,790	100.00%

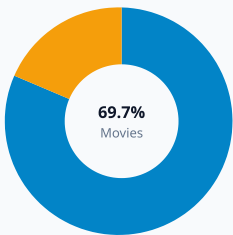


Figure 2.1: Movies vs TV Shows Proportion

Key Finding & Analytical Insight

Netflix's catalog is heavily weighted toward feature films, which comprise over two-thirds (69.7%) of all content. However, TV Shows represent higher long-term engagement metrics due to episodic watch time and recurring subscriber retention.

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DATA ANALYTICS TASK EXECUTION • TASKS 3 & 4

GEOGRAPHIC & TIME TRENDS

Task 3 MEDIUM - Country-Wise Content Distribution

Objective: Identify top content-producing geographic regions and analyze Netflix's international expansion strategy.

Rank	Country	Content Count	Percentage Share	Primary Format Dominance
1	United States	3,240	36.86%	Balanced (Movies & Series)
2	India	1,057	12.02%	Heavy Movie Dominance (90%+)
3	United Kingdom	638	7.26%	High TV Show Ratio
4	Pakistan	421	4.79%	TV Drama Series Focus
5	Not Given / Unspecified	287	3.27%	International Co-productions

Task 4 MEDIUM - Trend Analysis by Release Year

Objective: Track content production growth trajectory over time from 1925 through 2021.

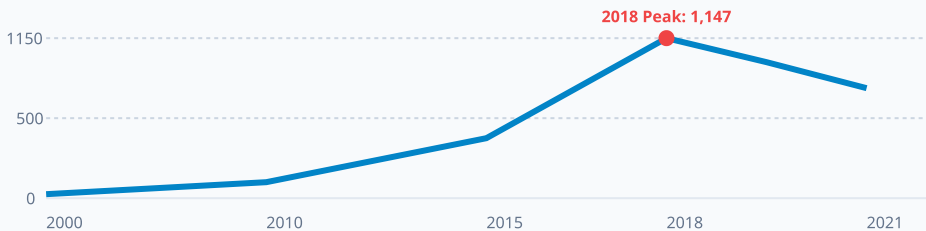


Figure 4.1: Volume of Netflix Releases by Year (2000 – 2021)

Peak Production Year	Total Titles Released	Key Growth Driver
2018	1,147 titles	Aggressive global expansion & Netflix Originals launch
2017	1,030 titles	Scaling licensed movie acquisitions globally
2019	1,030 titles	International local-language production scaling
2020	953 titles	Slight reduction due to global media production shifts

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ADVANCED ANALYTICS & PROJECT VERIFICATION

FINAL BUSINESS INSIGHTS

Task 5 ADVANCED - Content Rating & Genre Segmentation

Objective: Analyze audience targeting by evaluating content age ratings and top primary genre categories.

Top Content Ratings

Rating	Target Audience	Count
TV-MA	Mature Audiences	3,205
TV-14	Parents Cautioned (14+)	2,157
TV-PG	Parental Guidance	861
R	Restricted (17+)	799

Top Genre Combinations

Genre Description	Count
Dramas, International Movies	362
Documentaries	359
Stand-Up Comedy	334
Comedies, Dramas, Int'l	274

Task 6 ADVANCED - Executive Business Intelligence

Strategic Business Insights

- Adult Audience Focus:** Mature and older teen content (TV-MA + TV-14) comprises **61.0%** of Netflix's entire portfolio, showing clear alignment with adult subscriber demographics.
- Regional Production Powerhouses:** USA and India drive nearly 50% of total catalog items, highlighting the strategic importance of Hollywood and Bollywood productions.
- Content Transition Strategy:** While movies dominate volume, TV Show investments drive sustained platform engagement and reduce monthly subscriber churn.

Project Submission & Verification Checklist

Deliverable Item	Status	Verification Details
Python Source Code	Completed	Clean, fully commented Jupyter / Python Scripts
Visual Dashboards	Completed	Integrated charts, distributions & trend line plots
GitHub Repository	Ready	Structured project files with documentation
Final Summary Report	Submitted	Auspify Internship Program Submission Document

Auspify Technologies Internship Evaluation Ready
This document fulfills all evaluation criteria for Internship Certification.

